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Hou et al.

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(54) **ELECTRONIC DEVICE, WIRELESS COMMUNICATION METHOD, AND COMPUTER READABLE MEDIUM**

(58) **Field of Classification Search**
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H04L 1/1812; H04L 1/1678; H04L 1/16;
H04L 1/18

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(Continued)

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(73) Assignee: **SONY GROUP CORPORATION**, Tokyo (JP)

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(Continued)

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(57) **ABSTRACT**

The present invention relates to an electronic device, a wireless communication method, and a computer readable medium. According to one embodiment, the electronic device used for wireless communication comprises a processing circuit. The processing circuit is configured to perform control to send data and information related to retransmission of the data to a target user equipment by means of a sidelink. The processing circuit is further configured to control operation for the retransmission of the data.

(51) **Int. Cl.**

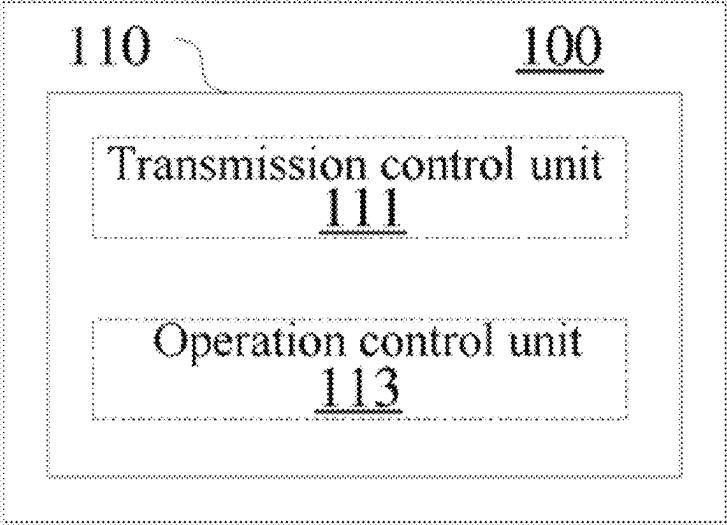
H04J 3/00 (2006.01)
H04L 1/1607 (2023.01)

(Continued)

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23 Claims, 13 Drawing Sheets



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USPC 370/329, 336
See application file for complete search history.
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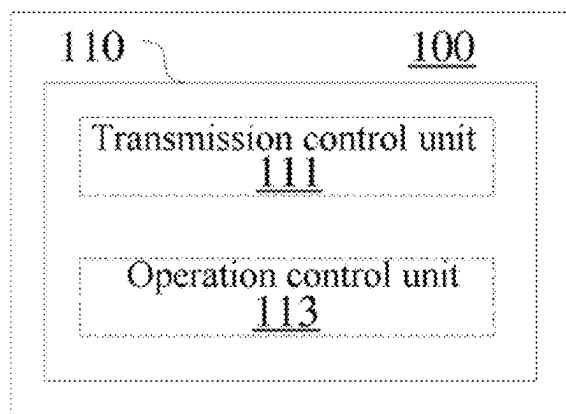


Figure 1

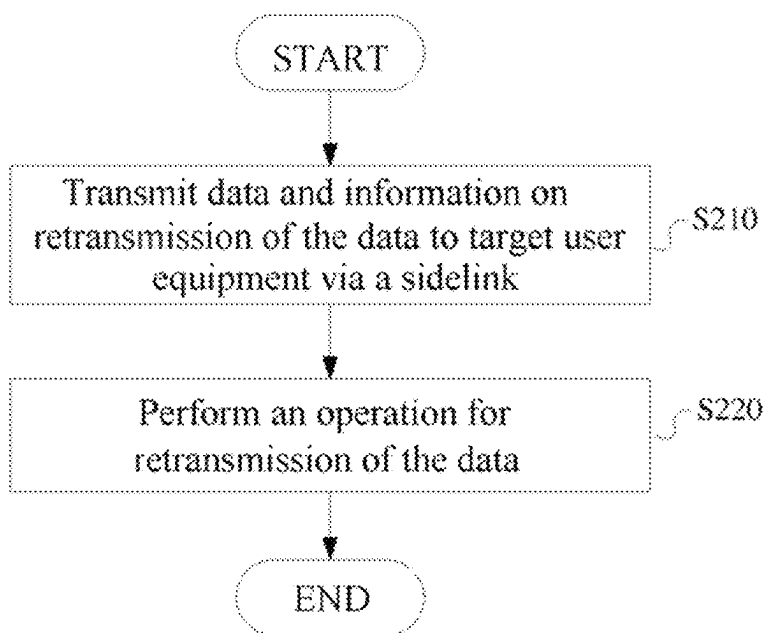


Figure 2

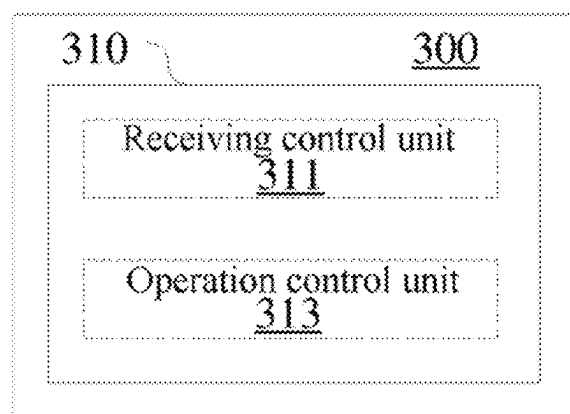


Figure 3

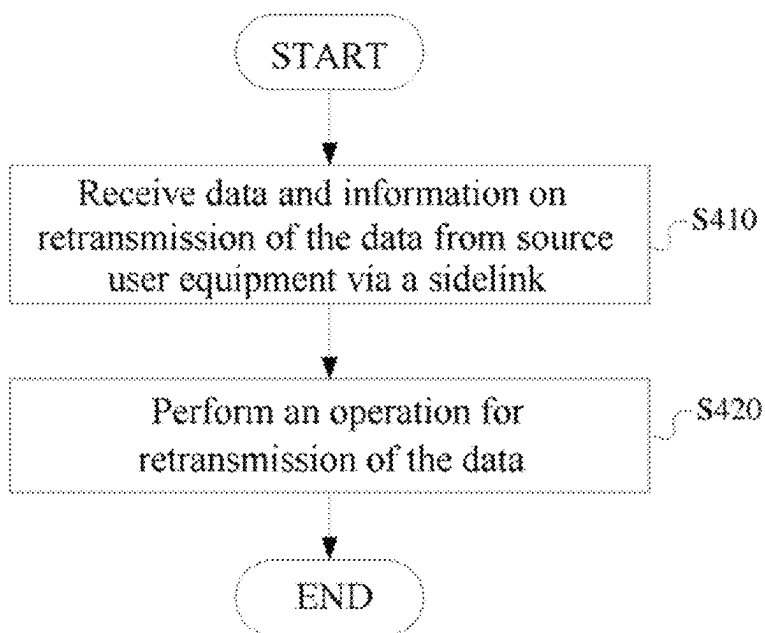


Figure 4

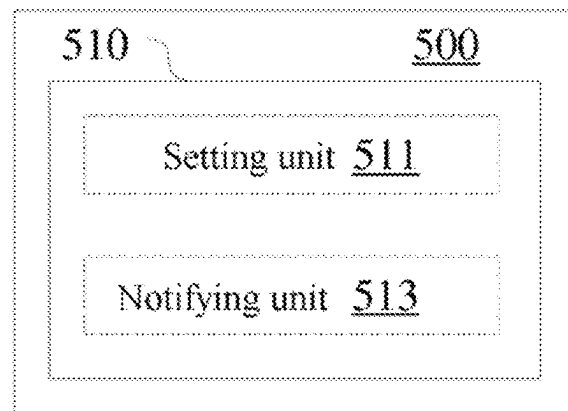


Figure 5

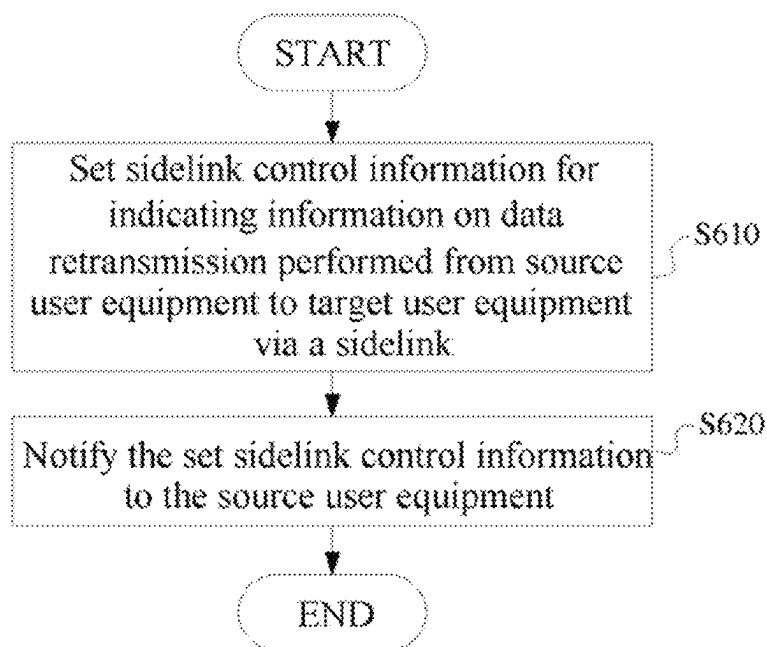


Figure 6

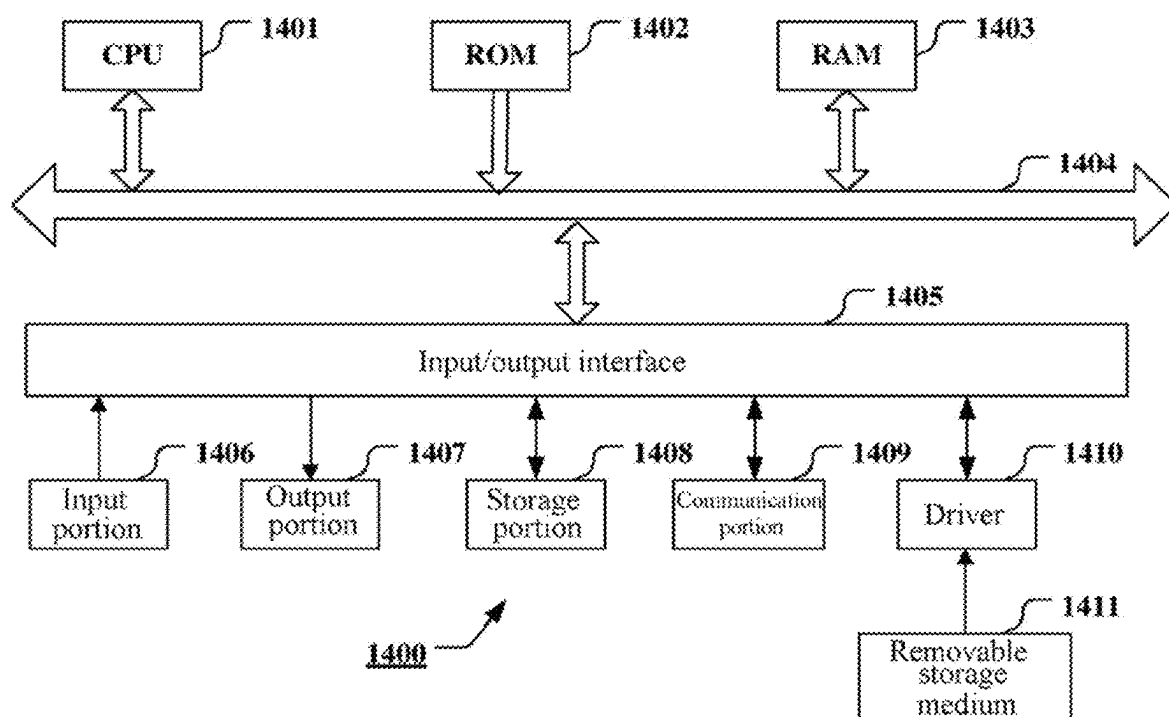


Figure 7

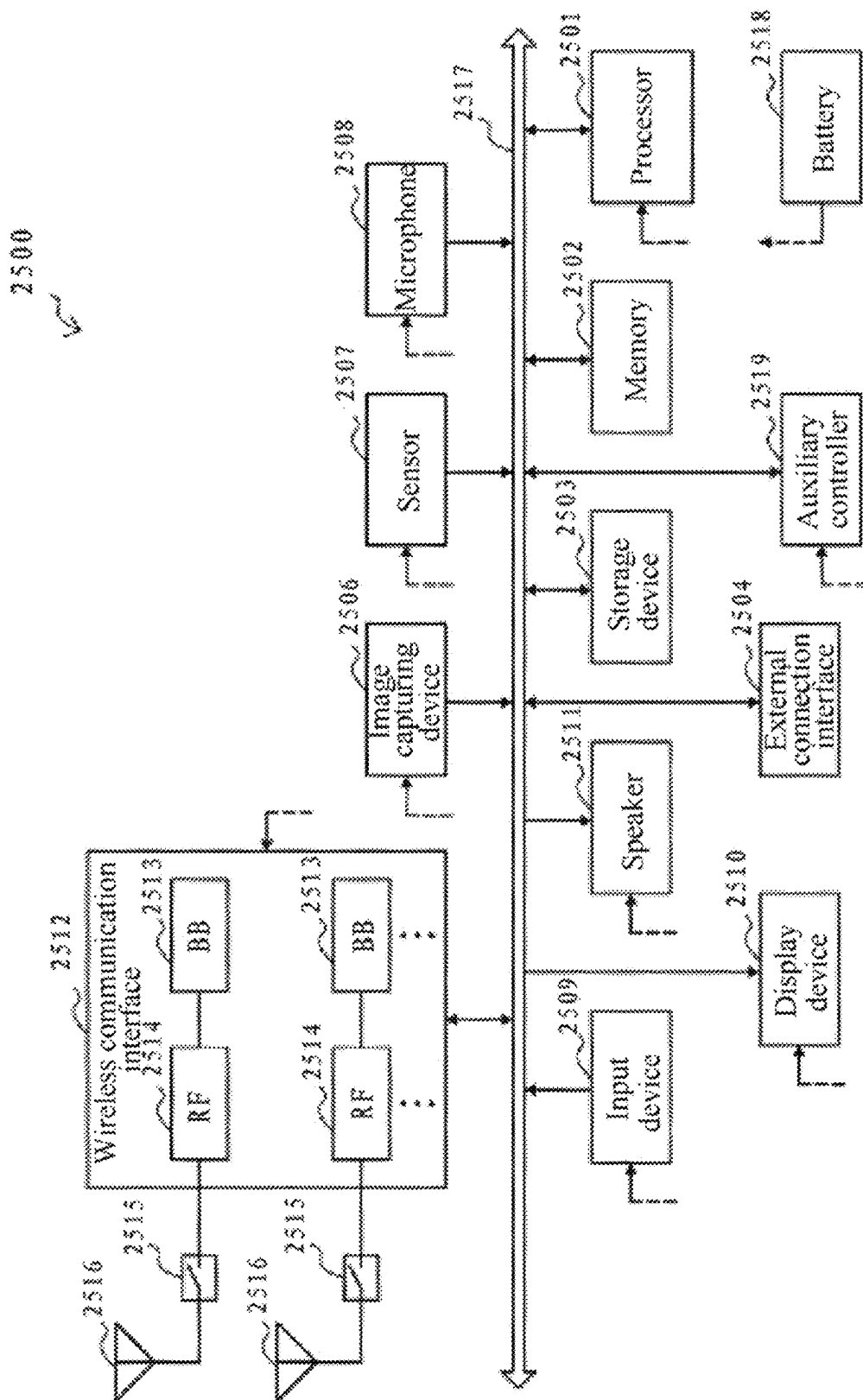


Figure 8

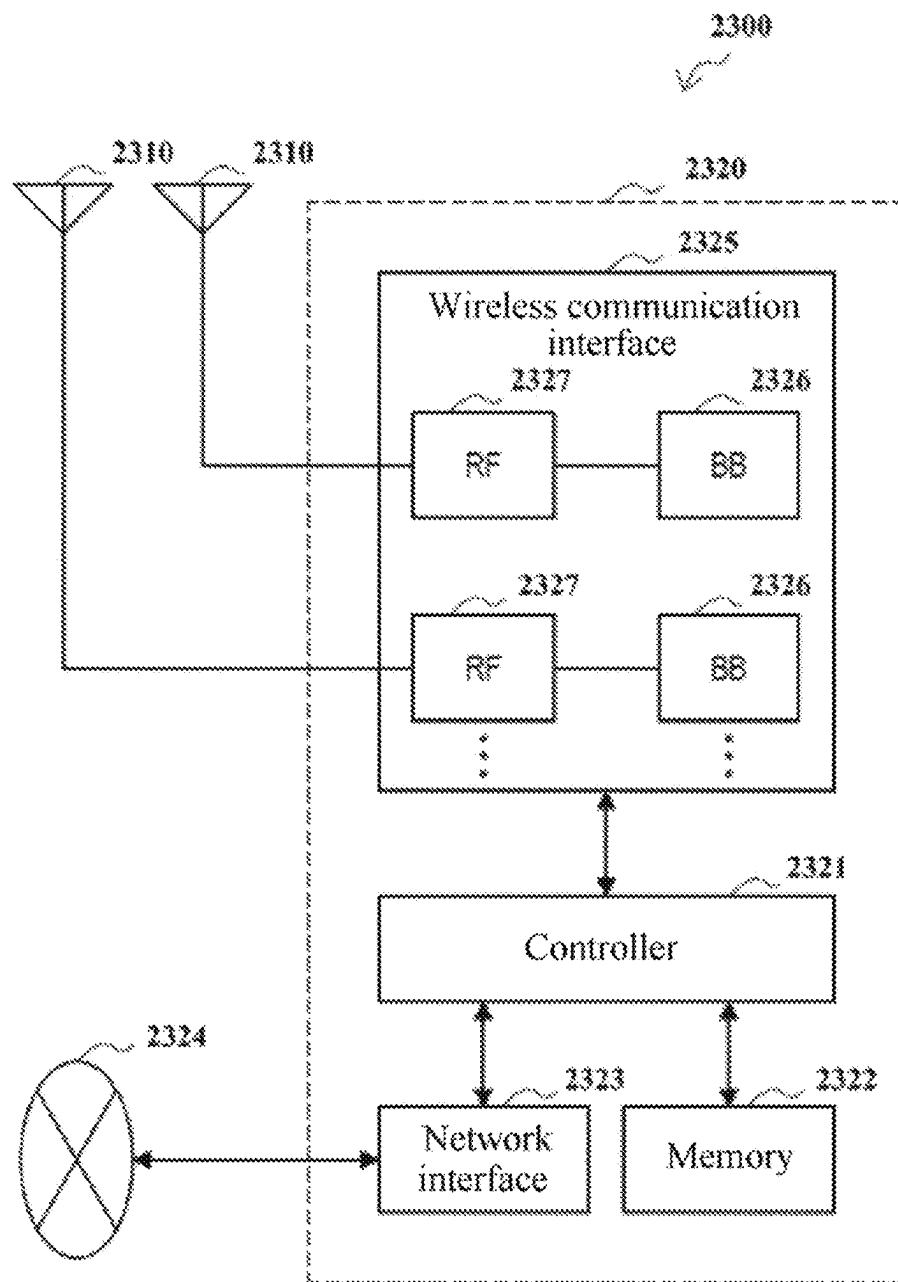


Figure 9

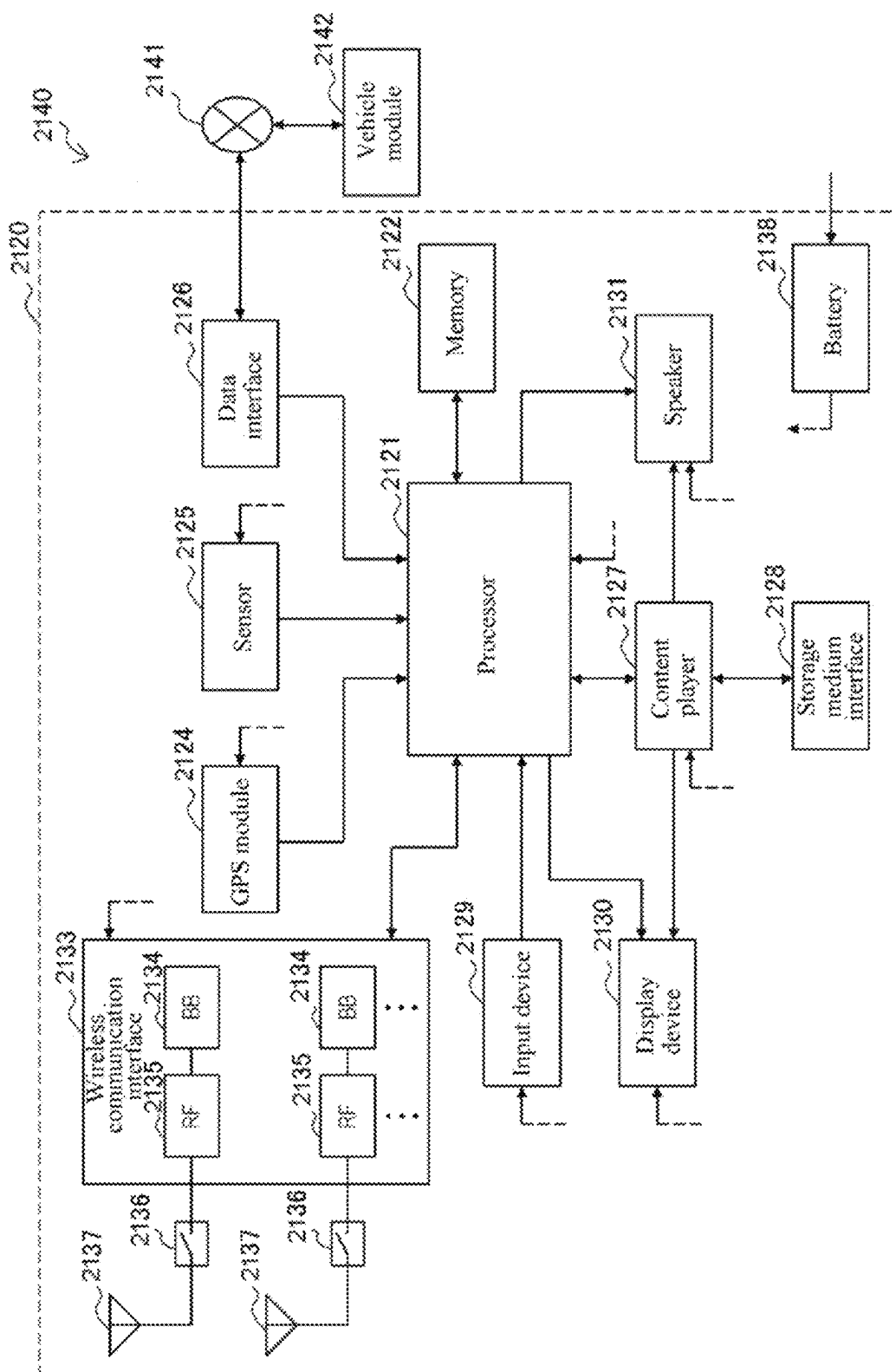


Figure 10

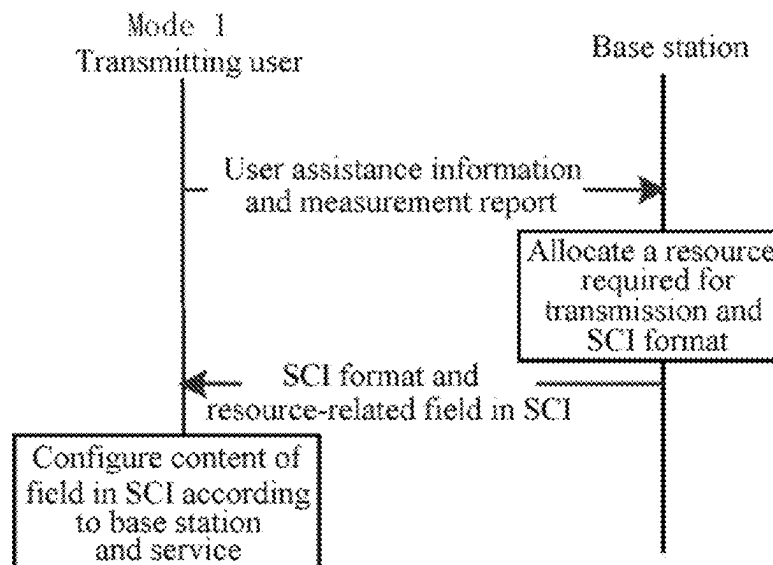


Figure 11

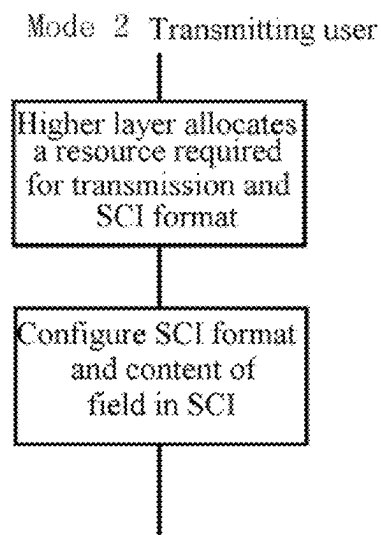


Figure 12

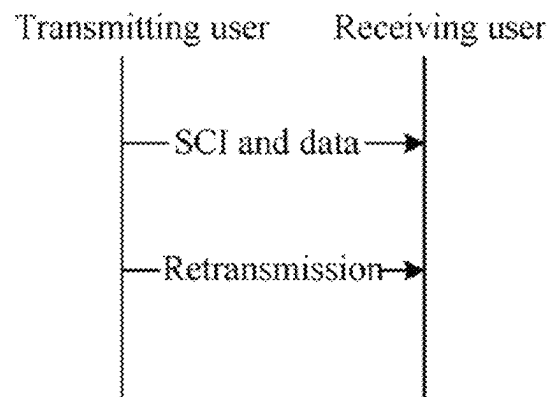


Figure 13

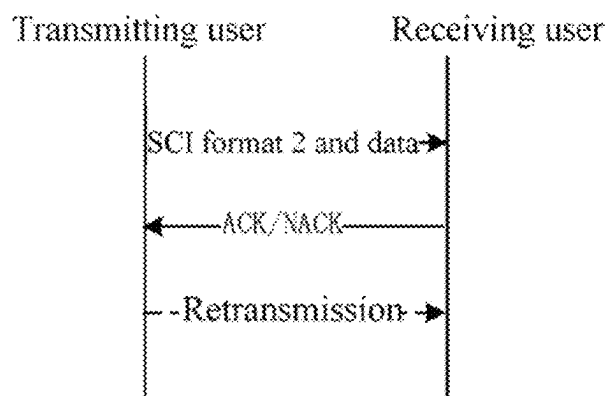


Figure 14

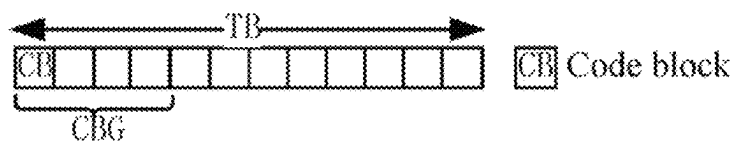


Figure 15

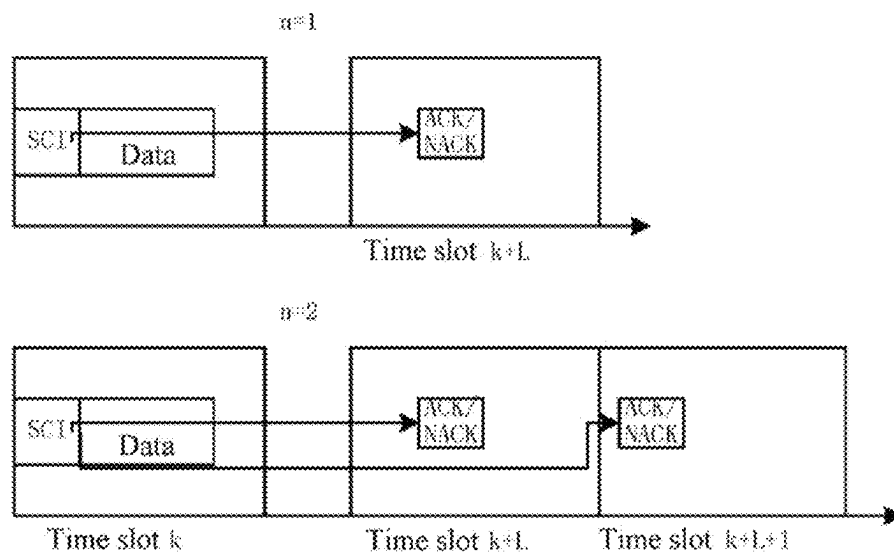


Figure 16

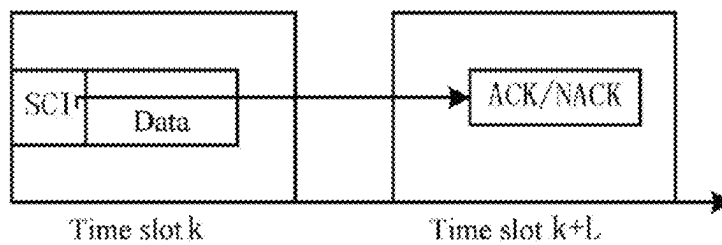


Figure 17

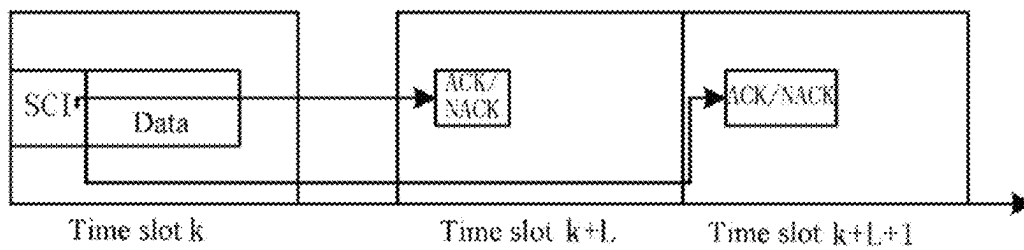


Figure 18

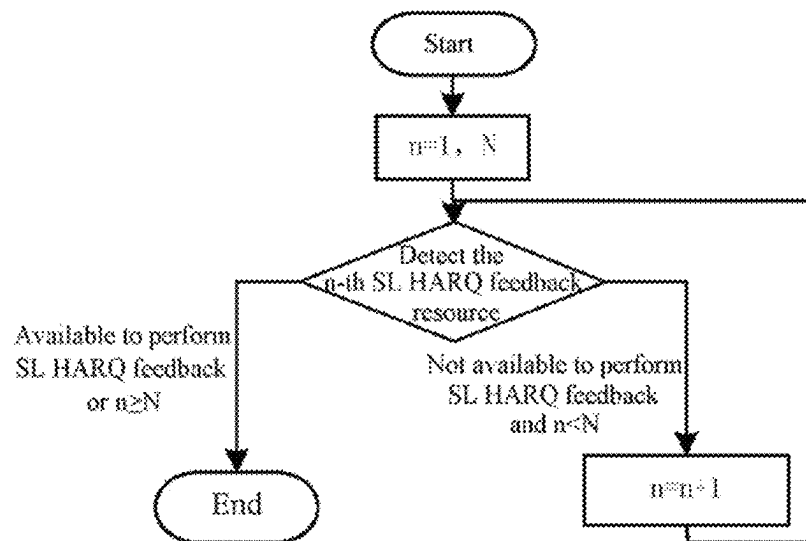


Figure 19

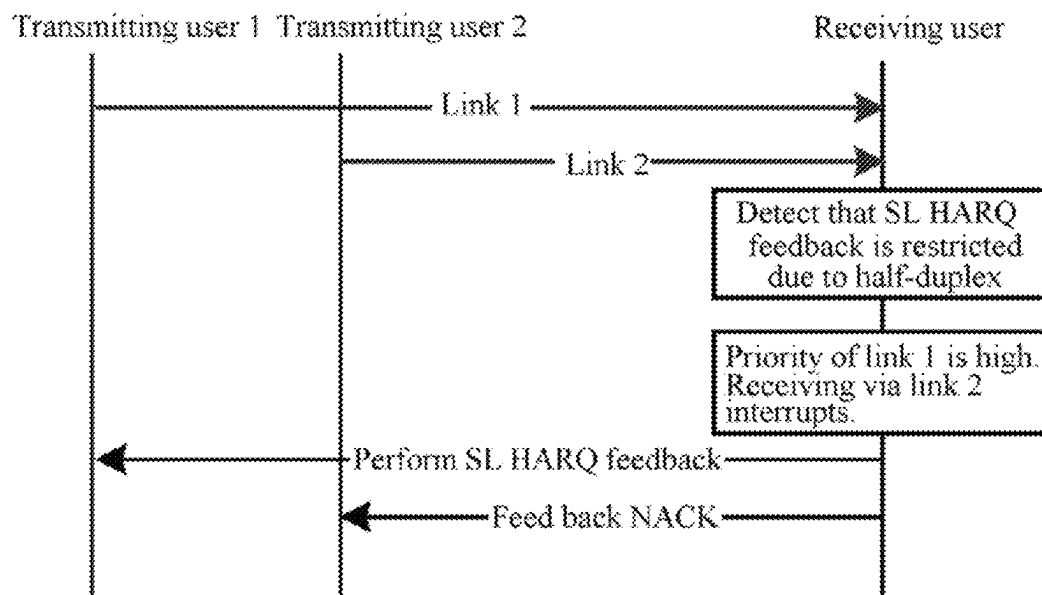


Figure 20

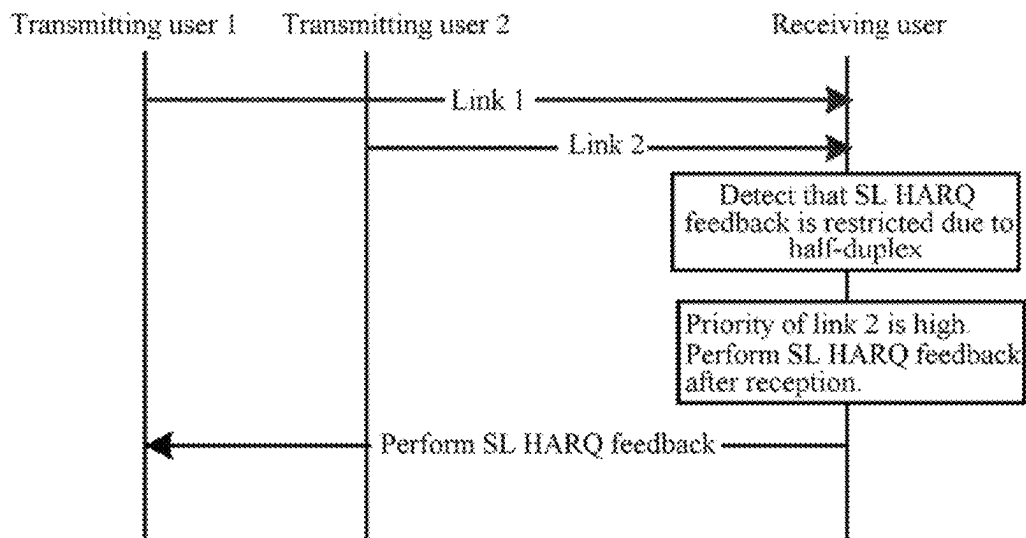


Figure 21

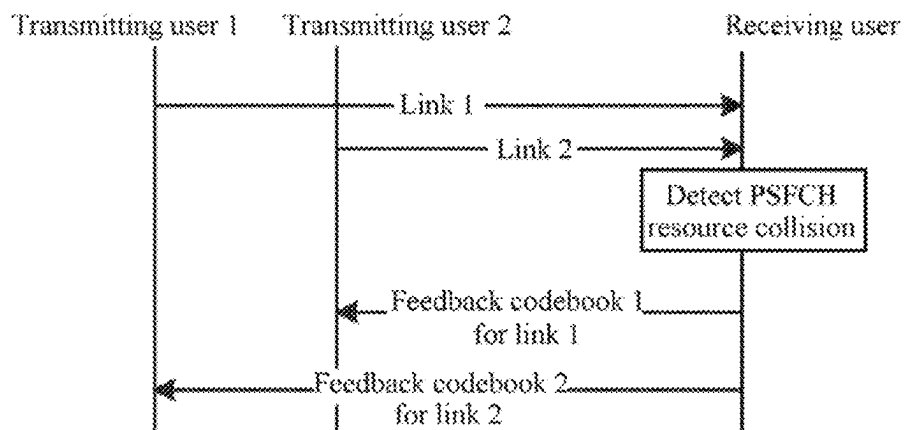


Figure 22

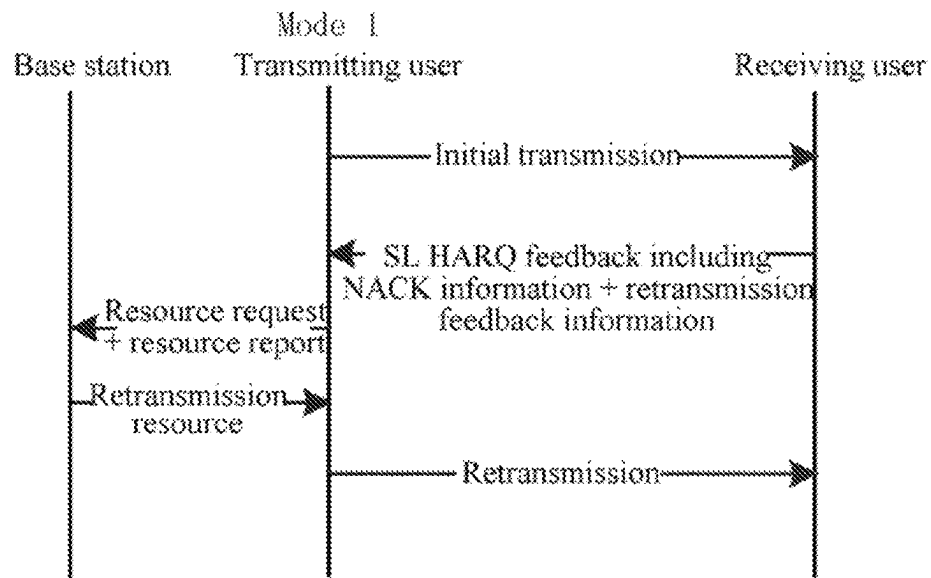


Figure 23

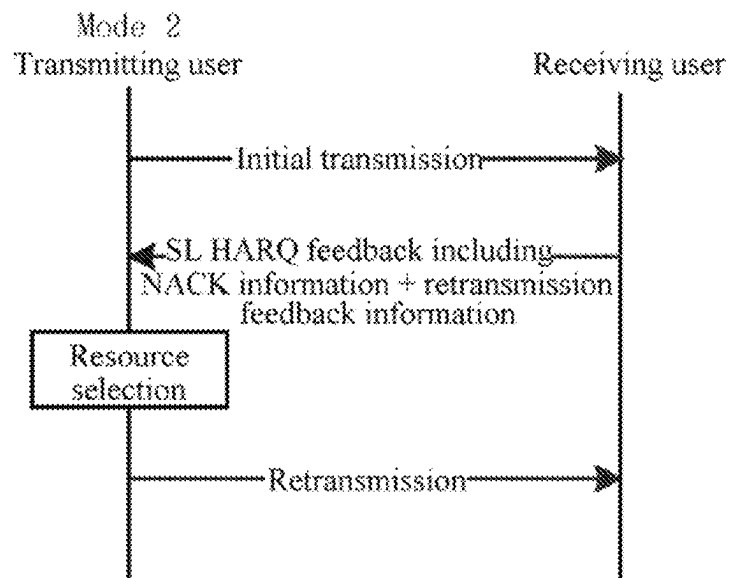


Figure 24

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ELECTRONIC DEVICE, WIRELESS COMMUNICATION METHOD, AND COMPUTER READABLE MEDIUM

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 17/424,167, filed on Jul. 20, 2021, which is based on PCT filing PCT/CN2020/074416, filed on Feb. 6, 2020, which claims the priority to Chinese Patent Application No. 201910112921.X, titled "ELECTRONIC DEVICE, WIRELESS COMMUNICATION METHOD, AND COMPUTER READABLE MEDIUM", filed on Feb. 13, 2019 with the China National Intellectual Property Administration (CNIPA), each of which is incorporated herein by reference in its entirety.

FIELD

The present disclosure generally relates to the field of wireless communication, and in particular to an electronic device for wireless communication, a wireless communication method, and a computer readable medium.

BACKGROUND

In TS 36.213 of 3GPP (Third Generation Partnership Project), user equipment (UE) procedures in LTE (Long Term Evolution) V2X (Vehicle-to-everything) communication are defined for determining time-frequency resources for UE to transmit PSCCH (Physical Sidelink Control Channel) and corresponding PSSCH (Physical Sidelink Shared Channel) and receiving the PSCCH, and information domain and configuration method of SCI (sidelink control information) are further defined. TS 36.321 defines a retransmission-based sidelink HARQ (Hybrid Automatic Repeat Request) process at the MAC (Media Access Control) layer in LTE V2X. TS 36.300 defines multi-link transmission in LTE V2X.

SUMMARY

In the following, an overview of the embodiments of the present disclosure is given simply to provide basic understanding to some aspects of the present disclosure. It should be understood that, the summary is not an exhaustive summary of the present disclosure. It is neither intended to determine a critical part or an important part of the present disclosure, nor to limit the scope of the present disclosure. The purpose is only to provide some concepts in a simplified form, as a preamble of a detailed description later.

According to an embodiment, an electronic device for wireless communication is provided. The electronic device includes processing circuitry. The processing circuitry is configured to perform control to transmit data and information on retransmission of the data to target user equipment via a sidelink. The processing circuitry is further configured to control an operation for retransmission of the data.

According to an embodiment, a wireless communication method includes a step of transmitting data and information on retransmission of the data to target user equipment via a sidelink. The method further includes a step of performing an operation for retransmission of the data.

According to an embodiment, an electronic device for wireless communication is provided. The electronic device includes processing circuitry. The processing circuitry is

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configured to perform control to receive data and information on retransmission of the data from source user equipment via a sidelink. The processing circuitry is further configured to control an operation for retransmission of the data.

According to an embodiment, a wireless communication method includes a step of receiving data and information on retransmission of the data from source user equipment via a sidelink. The method further includes a step of performing an operation for retransmission of the data.

According to an embodiment, an electronic device for wireless communication is provided. The electronic device includes processing circuitry. The processing circuitry is configured to set sidelink control information for indicating information on data retransmission performed from source user equipment to target user equipment via a sidelink. The processing circuitry is further configured to perform control to notify the set sidelink control information to the source user equipment. The information on data retransmission includes a time gap between initial transmission and retransmission of the data by the source user equipment, or time domain position information of a feedback to the data by the target user equipment.

According to an embodiment, a wireless communication method includes a step of setting sidelink control information. The sidelink control information is for indicating information on data retransmission performed from source user equipment to target user equipment via a sidelink. The method further includes a step of notifying the set sidelink control information to the source user equipment. The information on data retransmission includes a time gap between initial transmission and retransmission of the data by the source user equipment, or time domain position information of a feedback to the data by the target user equipment.

According to another embodiment, a computer-readable medium including executable instructions is provided. When executed by an information processing device, the executable instructions cause the information processing device to implement the above method.

With the embodiments of the present disclosure, data feedback and/or data retransmission in sidelink communication may be effectively performed, thereby ensuring reliability of the sidelink communication.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure may be understood better with reference to the description given in conjunction with drawings hereinafter. The same or similar reference numerals are used to indicate the same or similar components throughout the drawings. The drawings together with the following detailed description are included in the specification and form a part of the specification, so as to illustrate preferred embodiments of the present disclosure by examples and explain principles and advantages of the present disclosure. In the drawings:

FIG. 1 is a block diagram showing a configuration example of an electronic device for wireless communication according to an embodiment of the present disclosure;

FIG. 2 is a flowchart showing a process example of a wireless communication method according to an embodiment of the present disclosure;

FIG. 3 is a block diagram showing a configuration example of an electronic device for wireless communication according to another embodiment of the present disclosure;

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FIG. 4 is a flowchart showing a process example of a wireless communication method according to another embodiment of the present disclosure;

FIG. 5 is a block diagram showing a configuration example of an electronic device for wireless communication according to yet another embodiment of the present disclosure;

FIG. 6 is a flowchart showing a process example of a wireless communication method according to yet another embodiment of the present disclosure;

FIG. 7 is a block diagram showing an exemplary structure of a computer for implementing the method and the device according to the present disclosure;

FIG. 8 is a block diagram showing a schematic configuration of a smart phone to which the technology according to the present disclosure may be applied;

FIG. 9 is a block diagram showing an example of a schematic configuration of a gNB to which the technology according to the present disclosure may be applied;

FIG. 10 is a block diagram showing an example of a schematic configuration of a car navigation device to which the technology of the present disclosure may be applied;

FIG. 11 is a flowchart showing an example of a configuration of sidelink control information;

FIG. 12 is a flowchart showing another example of a configuration of sidelink control information;

FIG. 13 is a flowchart showing an example of a retransmission-based sidelink data transmission;

FIG. 14 is a flowchart showing an example of a feedback-based sidelink data transmission;

FIG. 15 is a schematic diagram showing a sidelink data transmission based on code block groups;

FIG. 16 is a schematic diagram illustrating an example of a time-frequency resource of sidelink feedback information;

FIG. 17 is a schematic diagram illustrating another example of a time-frequency resource of sidelink feedback information;

FIG. 18 is a schematic diagram illustrating another example of a time-frequency resource of sidelink feedback information;

FIG. 19 is a flowchart illustrating an example of feedback processing on a receiving user side;

FIG. 20 is a flowchart illustrating an example of feedback processing in a case of multiple transmitting users;

FIG. 21 is a flowchart illustrating another example of feedback processing in a case of multiple transmitting users;

FIG. 22 is a flowchart illustrating yet another example of feedback processing in a case of multiple transmitting users;

FIG. 23 is a flowchart illustrating an example of a process related to retransmission resource configuration; and

FIG. 24 is a flowchart illustrating an example of a process related to retransmission resource selection.

DETAILED DESCRIPTION

Embodiments of the present disclosure are described below with reference to the drawings. Elements and features described in one of the drawings or one of the embodiments of the present disclosure may be combined with elements and features described in one or more other drawings or embodiments. It should be noted that representations and descriptions of components and processing which are irrelevant to the present disclosure or known by those skilled in the art are omitted in the drawings and the description for clarity.

An exemplary configuration of an electronic device for wireless communication according to an embodiment of the

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present disclosure is described below with reference to FIG. 1. As shown in FIG. 1, an electronic device 100 for wireless communication according to the embodiment includes processing circuitry 110. The processing circuitry 110 may be implemented as, for example, a chip, a chipset, a central processing unit (CPU), or the like.

The electronic device according to the embodiment may be implemented on a user equipment side. More specifically, the user equipment may include a vehicle, for example. However, the present invention is not limited thereto, but may be used in various application scenarios of the New Wireless (NR) Sidelink, such as Machine-Type Communication (MTC), Device-to-Device (D2D) communication, Vehicle-to-everything (V2X) communication, Internet of Things (IOT) communication, drone communication, and the like. The V2X communication may include vehicle-to-vehicle (V2V) communication, vehicle-to-pedestrian (V2P) communication, vehicle-to-infrastructure (V2I) communication, and the like.

As shown in FIG. 1, the processing circuitry 110 includes a transmission control unit 111 and an operation control unit 113. It should be noted that, although the transmission control unit 111 and the operation control unit 113 are shown in a form of functional blocks in the drawings, it should be understood that functions of these units may also be implemented by the processing circuitry as a whole, and are unnecessarily implemented by discrete actual components in the processing circuitry. In addition, although the processing circuitry is shown as one block in the drawings, the electronic device may include multiple processing circuitry. The functions of these units may be distributed to the multiple processing circuitry, so that the multiple processing circuitry cooperate to perform these functions.

The transmission control unit 111 is configured to perform control to transmit data and information on retransmission of the data to target user equipment via a sidelink.

That is, the electronic device according to the embodiment may be implemented on a side of source user equipment (that is, the user equipment that operates as a transmission source of the transmitted data, and may also be referred to as a transmitting user in the present disclosure), and it may transmit the data to a side of target user equipment (that is, the user equipment that operates as a retransmission target of the transmitted data, and may also be referred to as a receiving user in the present disclosure).

The source user equipment may, for example, transmit information on retransmission of the data as sidelink control information (SCI) to the target user equipment.

The operation control unit 113 is configured to control an operation for retransmission of the data.

As an example, the operation for retransmission of the data may include an operation of a retransmission-based sidelink hybrid automatic repeat request (SL HARQ) or an operation of a feedback-based SL HARQ. The retransmission-based SL HARQ refers to a method in which the transmitting user performs multiple retransmissions to ensure reliability of communication. The feedback-based SL HARQ is a method in which the receiving user feeds back acknowledgment (ACK) information or non-acknowledgment (NACK) information to the transmitting user via a physical sidelink feedback channel (PSFCH), for example, to determine whether the transmitting user needs retransmission to ensure the reliability of communication.

Corresponding to the retransmission-based SL HARQ, according to an embodiment, the information on retransmission of data may include a time gap between initial transmission of the data and retransmission of the data. Corre-

spondingly, the operation for retransmission of the data may include performing retransmission of the data based on the time gap.

FIG. 13 shows an exemplary method for the retransmission-based SL HARQ. As shown in FIG. 13, a transmitting user transmits SCI and data to a receiving user, and performs retransmission of the data after a time gap between initial transmission and retransmission.

In addition to the time gap between initial transmission and retransmission, the SCI may further include one or more of the following information: service priority, resource reservation, source ID information, destination ID information, frequency domain position(s) for initial transmission and retransmission, modulation and coding scheme(s), retransmission index, and the like.

Regarding a field of the frequency domain position(s) for initial transmission and retransmission in the SCI, a Mode 1 (a mode involving base station assistance) user may configure content of the field based on a base station instruction, and a Mode 2 (a mode involving no base station assistance) user may configure content of the field based on a high-level instruction.

Similarly, regarding to a field of the time gap between initial transmission and retransmission in the SCI, the Mode 1 user may configure content of the field based on a base station instruction, and the Mode 2 user may configure content of the field based on a high-level instruction. In particular, for example, in a case where the field of the time gap between initial transmission and retransmission is configured as 0, the user may transmit a service without retransmission.

Corresponding to the feedback-based SL HARQ, according to another embodiment, the information on retransmission of the data may include time domain position information for feedback of the target user equipment to the data. Correspondingly, the operation for retransmission of the data may include receiving feedback information of the target user equipment to the data based on the time domain position information, and performing retransmission of the data based on the feedback information.

FIG. 14 shows an exemplary method for the feedback-based SL HARQ. As shown in FIG. 14, if the receiving user receives and successfully decodes a relevant transmission block (TB), the receiving user feeds back ACK information to the transmitting user. If the receiving user fails to decode the relevant transmission block after receiving it, the receiving user feeds back NACK information to the transmitting user, and the transmitting user performs retransmission after receiving the NACK feedback.

Although feedback and retransmission are performed in units of transmission blocks in this example, the present disclosure is not limited thereto. For example, the feedback and receiving may be performed in units of code block groups (CBG).

FIG. 15 is a schematic diagram showing a sidelink data transmission based on code block groups. As shown in FIG. 15, a TB contains multiple code blocks (CB), and several code blocks may form a code block group.

According to an embodiment, the information on retransmission of data may further include information indicating a retransmission unit (transmission block or code block group), and the operation for retransmission of data may include receiving a feedback to the data and performing retransmission of the data in units of transmission block or code block group.

More specifically, in a case of receiving feedback to the data and performing retransmission of the data in units of

code block group, the receiving user may generate 1-bit ACK information for a CBG if it receives and successfully decodes codeword in the CBG; and the receiving user may generate 1-bit NACK information for a CBG if it fails to receive and decode the codeword in the CBG.

If each TB contains N code block groups, the codebook may include N-bits ACK/NACK information. The receiving user feeds back the information of the codebook to the transmitting user.

Correspondingly, the SCI may include, for example, service priority, resource reservation, source ID information, destination ID information, frequency domain position(s) for initial transmission and retransmission, time domain position(s) for initial transmission and retransmission, modulation and coding scheme(s), retransmission index, HARQ ID, CBG transmission information, and the like.

The CBG transmission information may be used to indicate the unit (TB or CBG) for the receiving user to perform SL HARQ feedback. For example, if a CBG field is empty, it indicates a non-CBG SL HARQ, and if the CBG field is not empty, it indicates a CBG SL HARQ.

In addition, in unicast communication and multicast communication, if the communicating users all support the feedback-based SL HARQ, the receiving user feeds back ACK/NACK information to the transmitting user via the physical sidelink feedback channel (PSFCH). Regarding resource configuration of the PSFCH, the SCI sent from the transmitting user to the receiving user may include a frequency domain position of a feedback transmission and a time domain position of the feedback transmission.

According to an embodiment, the time domain position information may include one or more time domain positions for receiving feedback information. In addition, the time domain position information may include one or more time windows for receiving feedback information.

Next, an exemplary manner in which the transmitting user configures PSFCH resources for the receiving user in the non-CBG SL HARQ is explained. The transmitting user indicates a time-frequency resource for sidelink feedback information to be used by the receiving user to feed back ACK/NACK information.

As shown in FIG. 16, according to an exemplary embodiment, in the SCI, the transmitting user indicates n (n is greater than or equal to 1) time domain and frequency domain positions for the sidelink feedback information.

As shown in FIG. 17, according to another exemplary embodiment, in the SCI, the transmitting user indicates a frequency domain position and a feedback time window for the sidelink feedback information.

As shown in FIG. 18, according to yet another exemplary embodiment, in the SCI, the transmitting user indicates a time-frequency position and a feedback time window for the sidelink feedback information.

In addition, for the CBG SL HARQ, the transmitting user may configure PSFCH resources for the receiving user in a similar way. The difference is that the time-frequency resource indicated by the transmitting user for the sidelink feedback information may be used for feedback codebook information.

After receiving the SCI information and decoding based on indications in the SCI field, the receiving user may determine a SL HARQ feedback resource. The transmitting user waits to receive the SL HARQ feedback information at the PSFCH resource position configured for the receiving user.

Next, exemplary methods for SL HARQ feedback are explained respectively corresponding to the foregoing feedback resource configurations.

According to an embodiment, the time domain position information for data retransmission may include multiple time domain positions for receiving feedback information, and the operation for retransmission of the data may include: receiving the feedback information successively at the multiple time domain positions until the feedback information is received; and performing retransmission if the feedback information is not received at any of the multiple time domain positions.

FIG. 19 shows an exemplary process, where N represents the number of resources configured for the SL HARQ feedback information, and the SL HARQ feedback resources are referred to, in chronological order, as a first SL HARQ resource, a second SL HARQ resource, and the like.

If the receiving user determines through detection that the first SL HARQ feedback resource configured by the transmitting user is not occupied and the time domain position thereof is available for transmission, the receiving user transmits the SL HARQ feedback information over the first SL HARQ feedback resource. Otherwise, the receiving user continues to detect the next feedback resource.

If none of the N feedback resources indicated by the transmitting user is available to perform SL HARQ feedback, the receiving user does not provide feedback to this transmission.

Correspondingly, in a case where the transmitting user receives the SL HARQ feedback information at the first SL HARQ resource position, the transmission is finished if the transmitting user receives ACK information, and retransmission is performed if the transmitting user receives NACK information.

If the transmitting user fails to receive the SL HARQ feedback information at the first SL HARQ resource position, the transmitting user continues to wait to receive the SL HARQ feedback at the next SL HARQ resource position. If the SL HARQ is received, the feedback is received successfully; otherwise, the steps are performed until the feedback information is received at one of the positions of the SL HARQ feedback resources.

If the transmitting user fails to receive the SL HARQ feedback at any of the positions of PSFCH resources configured for the receiving user, the retransmission is performed.

According to another embodiment, the time domain position information for data retransmission may include multiple time windows for receiving feedback information, and the operation for retransmission of the data may include: receiving the feedback information successively within the multiple time windows until the feedback information is received; and performing retransmission if the feedback information is not received within any of the multiple time windows.

More specifically, for example, the receiving user detects, within the feedback time window configured by the transmitting user, a time-frequency resource that is available to perform the SL HARQ feedback, and selects a time domain resource for the SL HARQ feedback based on a time delay requirement by a service, and performs the SL HARQ feedback. If no time-frequency resource that is available to perform the SL HARQ feedback is detected, the receiving user does not provide feedback to this transmission.

Correspondingly, the transmitting user waits to receive, within the time window of the PSFCH configured for the receiving user, the SL HARQ feedback. If the transmitting

user receives the SL HARQ feedback information, the feedback is received successfully. If the transmitting user fails to receive the SL HARQ feedback at a resource part within the PSFCH time window, the retransmission step is performed.

According to another embodiment, the time domain position information for data retransmission includes a time domain position and a time window for receiving feedback information, and the operation for data retransmission may include: receiving the feedback information successively at the time domain position and within the time window, until the feedback information is received; and performing retransmission if failing to receive the feedback information at the time domain position or within the time window.

More specifically, if the receiving user detects that the first SL HARQ feedback resource configured by the transmitting user is not occupied and the time domain position thereof is available for transmission, the receiving user transmits the SL HARQ feedback information over the first SL HARQ feedback resource.

If the receiving user detects that the first SL HARQ feedback resource configured by the transmitting user is occupied or the time domain position thereof is not available for transmission, and an SL HARQ feedback time window is indicated by the transmitting user, the receiving user continues to detect the resources within the SL HARQ feedback time window, selects the time domain resource for the SL HARQ feedback based on a time delay requirement by a service, and performs the SL HARQ feedback.

If no resource is available for the SL HARQ feedback, the receiving user does not provide feedback to this transmission.

Correspondingly, in a case where the transmitting user receives the SL HARQ feedback information at the first SL HARQ resource position, the transmission is finished if the transmitting user receives ACK information, and retransmission is performed if the transmitting user receives NACK information.

If the transmitting user fails to receive the SL HARQ feedback information at the first SL HARQ resource position, the transmitting user continues to wait to receive, within the SL HARQ resource time window, the SL HARQ feedback. If the transmitting user receives the SL HARQ, the feedback is received successfully; otherwise, the transmitting user performs retransmission.

Processes for feedback and retransmission are described in conjunction with specific examples. In addition, in a case where the transmitting user 1 and the transmitting user 2 select transmitting resources through resource sensing, and communicate with receiving user 3 via link 1 and link 2, respectively, PSFCH resources determined for the receiving user 3 by the transmitting user 1 and transmitting user 2 may overlap in whole or in part. An exemplary embodiment for a situation of feedback resource collision is explained below.

Referring to FIG. 22, when the receiving user detects that the PSFCH resources of link 1 and link 2 collide, the receiving user processes the SL HARQ feedback information of link 1 and the SL HARQ feedback information of link 2 and transmits the processed SL HARQ feedback information to the transmitting user 1 and the transmitting user 2, respectively.

The transmitting user may detect a signal on the PSFCH resource, and decode and receive the feedback signal using information on the transmitting user side. The transmitting user can only decode the feedback information on the link

where the transmitting user is located. Therefore, the problem of feedback resource collision may be solved.

Correspondingly, according to an embodiment, the operation for data retransmission on the transmitting user side may include decoding the received feedback information, and determining whether the decoded feedback information is for the data transmitted from the transmitting user.

More specifically, the processing of the SL HARQ feedback information may include transmitting the user's ID or HARQ ID through respective links. Alternatively, the processing may include scrambling by codebook or the like.

Correspondingly, according to an embodiment, the operation for data retransmission on the transmitting user side may include performing feedback to the data and retransmission of the data in units of code block group, and the feedback information may include codebook information.

In addition, for a case where the receiving user receives data through multiple links, the maximum number M of links for feedbacks supported on a same resource may be set in consideration of power limitation for the user. The data may be configured through system information or RRC (Radio Resource Control) signaling, selected based on position, or pre-configured, for example.

Assuming that the receiving user detects that a PSFCH collision occurs for m ($m > 1$) links: if $M \geq m$, for each of the links, the scheme in the foregoing embodiment for feedback is adopted. If $M < m$, for example, priorities of the links are compared, and feedback is performed for M links with the highest priority over this PSFCH resource, and for the remaining links, feedback may be performed over other feedback resources thereof, or no feedback may be performed if there is no additional feedback resources.

Exemplary manners for retransmission of data based on feedback are explained below.

For non-CBG SL HARQ, after receiving the NACK information fed back by the receiving user, the transmitting user may perform retransmission of relevant transmission block (TB) with respect to this HARQ. For CBG SL HARQ, after receiving the codebook information fed back by the receiving user, the transmitting user may perform retransmission of CBG whose feedback is NACK in the codebook.

The transmitting user may perform retransmission of the data through a same resource or a different resource from that for the initial transmission of the data. In addition, the transmitting user may determine the resource for data retransmission according to the feedback information from the receiving user.

For example, the receiving user may transmit the retransmission feedback information while transmitting the SL HARQ feedback information including the NACK information, so as to assist the transmitting user in selecting resources for SL HARQ retransmission. The retransmission feedback information may include, for example, a set of retransmission resource candidates, a measurement result of (part or all) resource sensing performed by the receiving user, channel busy ratio (CBR) measurement information, channel state information (CSI), and the like.

Correspondingly, according to an embodiment, the feedback information from the receiving user may include information on a retransmission resource for the data, and the operation for the retransmission of the data may include determining the retransmission resource for the data according to the information on retransmission resource.

In a case where the transmitting user is a Mode 1 user, based on a resource measurement report from the transmitting user and the retransmission feedback information from the receiving user, a base station may configure retransmis-

sion resource for the transmitting user according to a resource pool situation and the resource report. In a case where the transmitting user is a Mode 2 user, the transmitting user may select the retransmission resource based on the resource measurement report and the retransmission feedback information from the receiving user.

In addition, in data retransmission, for example, information such as retransmission index and HARQ ID may be set in the SCI in order to distinguish between initial transmission and retransmission.

FIG. 23 shows an exemplary process of a Mode 1 transmitting user obtains a retransmission resource according to base station configuration and resource report information.

First, the transmitting user receives and decodes SL HARQ feedback information and retransmission feedback information from the receiving user. The Mode 1 user then transmits a resource request and a resource report to the base station after receiving NACK information. The base station allocates a retransmission resource according to the retransmission request from the Mode 1 user, and notifies the same to the transmitting user. The Mode 1 user perform retransmission over the retransmission resource.

FIG. 24 shows an exemplary process of a Mode 2 transmitting user selecting a retransmission resource.

First, the transmitting user receives and decodes SL HARQ feedback information and retransmission feedback information from the receiving user. The Mode 2 user then selects a suitable retransmission resource according to the retransmission feedback request and the resource measurement result, and performs retransmission over the retransmission resource.

In the above description of the electronic device according to the embodiments of the present disclosure, some processes and methods are disclosed. Next, a wireless communication method according to an embodiment of the present disclosure is described without repeating the details described above.

As shown in FIG. 2, the wireless communication method according to the embodiment includes a step S210 of transmitting data and information on retransmission of the data to target user equipment via a sidelink, and a step S220 of performing an operation for retransmission of the data.

This method may be implemented on the transmitting user side described above.

In addition, a device and a method implemented for a receiving user side are further provided in the present disclosure. Next, a description of the embodiment for the device and the method implemented for a receiving user side is given without repeating the details corresponding to the above description of that for the transmitting user side.

As shown in FIG. 3, an electronic device 300 for wireless communication according to the embodiment includes processing circuitry 310. The processing circuitry 310 includes a receiving control unit 311 and an operation control unit 313.

The electronic device according to the embodiment may be implemented on a user equipment side such as a vehicle, and may be applied to the aforementioned application scenarios of the sidelink.

The receiving control unit 311 is configured to perform control to receive data and information on retransmission of the data from source user equipment via a sidelink.

The operation control unit 313 is configured to control an operation for retransmission of the data.

According to an embodiment, the information on retransmission of the data may include a time gap between initial

transmission and retransmission of the data, and the operation for the retransmission of the data may include receiving the retransmission of the data according to the time gap.

According to another embodiment, the information on retransmission of the data may include a time domain position information of a feedback to the data, and the operation for the retransmission of the data may include transmitting feedback information to the data to source user equipment based on the time domain position information.

The information on retransmission of the data may further include information indicating a retransmission unit, and the operation for the retransmission of the data may include performing a feedback to the data in units of transmission block or code block group.

The time domain position information may include one or more time domain positions for transmitting the feedback information and/or one or more time windows for transmitting the feedback information. The feedback information may be transmitted at one of the one or more time domain positions or within one of the one or more time windows.

According to an embodiment, the information on retransmission of the data may include a service priority of the data, and the operation for the retransmission of the data may include transmitting the feedback information and receiving other data based on the service priority, or may include transmitting the feedback information for multiple links.

More specifically, in a case where the number of the multiple links exceeds a predetermined threshold, the link for which the feedback information is to be transmitted may be selected according to the priority.

In addition, in a case where the receiving user detects that the SL HARQ feedback resource for the first link is restricted due to half-duplex because the second link is receiving information, the feedback of each link may also be performed according to the priority.

As shown in FIG. 20, in a case where the receiving user detects that the SL HARQ feedback is restricted due to half-duplex, the receiving user selects the link to be preferentially processed according to the service priority or QoS (Quality of Service) level of each of link 1 and link 2.

If the priority of link 1 is higher than the priority of link 2, the receiving user preferentially performs the SL HARQ feedback for link 1 and interrupts receiving information via link 2.

In a case where link 2 supports CBG SL HARQ, if the information can still be received via link 2 after the SL HARQ feedback is completed, the receiving user continues to receive the information via link 2. After the transmission via link 2 is completed, NACK information is fed back for CBG that is not received and retransmission is requested. In a case where link 2 supports non-CBG SL HARQ, if the information can still be received via link 2 after the SL HARQ feedback is completed, the receiving user continues to receive the information via link 2. After the transmission via link 2 is completed, NACK information is fed back to transmitting user 2.

On the other hand, if the priority of link 2 is higher than the priority of link 1, the receiving user preferentially receives information via link 2 rather than performing the SL HARQ feedback for link 1. After finishing receiving the information via link 2, the receiving user performs the SL HARQ feedback for link 1 according to the indication in the SCI, as shown in FIG. 21.

According to an embodiment, the feedback information may include identification information of corresponding source user equipment or may be scrambled through a

codebook in order to distinguish the feedback information for multiple links from each other.

In addition, it is also possible that the feedback information includes the information on a retransmission resource for the data, and the retransmission of the data is received based on the retransmission resource.

FIG. 4 shows an exemplary process of a corresponding wireless communication method. The method includes a step S410 of receiving data and information on retransmission of the data from source user equipment via a sidelink, and a step S420 of performing an operation for retransmission of the data.

In addition, a device and a method implemented for a base station side are further provided in the present disclosure. Next, a description of the embodiment for the device and the method implemented for a base station side is given without repeating the details corresponding to the above description of that for the user side.

As shown in FIG. 5, an electronic device 500 for wireless communication according to the embodiment includes processing circuitry 510. The processing circuitry 510 includes a setting unit 511 and a notifying unit 513.

The setting unit 511 is configured to set sidelink control information. The sidelink control information is for indicating information on data retransmission performed from source user equipment to target user equipment via a sidelink. The information on data retransmission includes a time gap between initial transmission and retransmission of the data by the source user equipment or time domain position information of a feedback to the data by the target user equipment.

In other words, the set SCI format may include a SCI format for retransmission-based SL HARQ and a SCI format for feedback-based SL HARQ.

In addition, user assistance information may be set, which may include, for example, the maximum transmission block size of service transmission data, service priority information, service SL HARQ mechanism preferences, and other information.

The notifying unit 513 is configured to perform control to notify the set sidelink control information to the source user equipment.

More specifically, as shown in FIG. 11, for a Mode 1 user, the base station may configure the SCI format and content of a transmission resource-related field in the SCI according to the user assistance information, channel measurement, and other information reported by the user.

For the Mode 1 user, the resource-related field in the SCI may include a time gap between initial transmission and retransmission, frequency domain positions of the initial transmission and the retransmission, or time domain position of the initial transmission, frequency domain position of the initial transmission, time domain indication of SL HARQ feedback, frequency domain indication of SL HARQ feedback, and other information.

In addition, as shown in FIG. 12, for a Mode 2 user, a higher layer of the user may configure the SCI format and the content of a transmission resource-related field in the SCI according to the user assistance information, channel measurement, and other information reported by the user.

FIG. 6 shows an exemplary process of a corresponding wireless communication method. The method includes a step S610 of setting sidelink control information. The sidelink control information is for indicating information on data retransmission performed from source user equipment to target user equipment via a sidelink. The information on data retransmission may include a time gap between initial

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transmission and retransmission of the data by the source user equipment, or time domain position information of a feedback to the data by the target user equipment. The method further includes a step **S620** of notifying the set sidelink control information to the source user equipment.

In addition, an embodiment of the present disclosure further includes a computer-readable medium, which includes executable instructions that, when executed by an information processing device, cause the information processing device to implement the above method.

The foregoing embodiment relates to a SL HARQ feedback mechanism when HARQ feedbacks are performed for multiple links based on a SL HARQ configuration and when a receiving user is restricted due to half-duplex, in order to ensure reliability of communication.

In addition, the foregoing embodiment also involves a selection of SL HARQ retransmission resources in order to ensure validity of a retransmission resource of a user.

As an example, steps of the above methods and modules and/or units of the above devices may be implemented as software, firmware, hardware or a combination thereof. In a case of implementing by software or firmware, programs constituting the software for implementing the above methods may be installed to a computer with a dedicated hardware structure (for example, a general-purpose computer **1400** shown in FIG. 7) from a storage medium or a network. The computer is capable of performing various functions when installed with various programs.

In FIG. 7, a central processing unit (CPU) **1401** performs various processing based on programs stored in a read only memory (ROM) **1402** or programs loaded from a storage portion **1408** to a random access memory (RAM) **1403**. Data required when the CPU **1401** performs various processing is also stored in the RAM **1403** as needed. The CPU **1401**, the ROM **1402**, and the RAM **1403** are linked to each other via a bus **1404**. An input/output interface **1405** is also linked to the bus **1404**.

The input/output interface **1405** is linked with an input portion **1406** (including a keyboard, a mouse and the like), an output portion **1407** (including a display, such as a Cathode Ray Tube (CRT) and a Liquid Crystal Display (LCD), and a loudspeaker), a storage portion **1408** (including a hard disk and the like), and a communication portion **1409** (including a network interface card, such as a LAN card, a modem and the like). The communication portion **1409** performs communication processing via a network such as the Internet. A driver **1410** may also be linked to the input/output interface **1405** as needed. A removable medium **1411** such as a magnetic disk, an optical disk, a magneto-optical disk, and a semiconductor memory is mounted on the driver **1410** as needed, so that a computer program read from the removable medium is installed into the storage portion **1408** as needed.

In a case where the above series of processing are implemented by software, programs forming the software are installed from a network such as the Internet or a storage medium such as the removable medium **1411**.

Those skilled in the art should understand that the storage medium is not limited to the removable medium **1411** shown in FIG. 7 that stores a program and is distributed separately from a device so as to provide the program to the user. Examples of the removable medium **1411** include: a magnetic disk (including a floppy disk (registered trademark)), an optical disk (including a compact disk read only memory (CD-ROM) and a digital versatile disk (DVD)), a magneto-optical disk (including a mini disk (MD) (registered trademark)), and a semiconductor memory. Alternatively, the

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storage medium may be the ROM **1402**, a hard disk included in the storage portion **1408** or the like. The storage medium has a program stored therein and is distributed to the user together with a device in which the storage medium is included.

A program product storing machine readable instruction codes is further provided according to an embodiment of the present disclosure. The instruction codes, when being read and executed by a machine, may perform the methods according to the above embodiments of the present disclosure.

Accordingly, a storage medium for carrying the program product storing the machine-readable instruction codes is also provided according to the present disclosure. The storage medium may include, but not limited to, a floppy disk, an optical disk, a magneto-optical disk, a memory card, a memory stick and the like.

The following electronic device is further involved in the embodiments of the present disclosure. In a case where the electronic device is used for a base station side, the electronic device may be implemented as any type of gNB or evolved node B (eNB), such as a macro eNB and a small eNB. The small eNB may be an eNB covering a cell smaller than a macro cell, such as a pico eNB, a micro eNB and a home (femto) eNB. Alternatively, the electronic device may be implemented as any other type of base station, such as a NodeB and a base transceiver station (BTS). The electronic device may include: a body configured to control wireless communications (which is also referred to as a base station equipment); and one or more remote radio heads (RRH) located at a different position from the body. In addition, various types of terminals, which are described below, may each serve as a base station by performing functions of the base station temporarily or semi-persistently.

In a case where the electronic device is used for a user equipment side, the electronic device may be implemented as a mobile terminal (such as a smartphone, a tablet personal computer (PC), a notebook PC, a portable game terminal, a portable/dongle mobile router and a digital camera) or a vehicle terminal (such as an a vehicle navigation device). Furthermore, the electronic device may be a wireless communication module (such as an integrated circuitry module including a single or multiple dies) mounted on each of the terminals described above.

[Application Example for a Terminal Equipment]

FIG. 8 is a block diagram showing an exemplary configuration of a smartphone **2500** to which the technology according to the present disclosure may be applied. The smartphone **2500** includes a processor **2501**, a memory **2502**, a storage device **2503**, an external connection interface **2504**, an image capturing device **2506**, a sensor **2507**, a microphone **2508**, an input device **2509**, a display device **2510**, a speaker **2511**, a wireless communication interface **2512**, one or more antenna switches **2515**, one or more antennas **2516**, a bus **2517**, a battery **2518** and an auxiliary controller **2519**.

The processor **2501** may be, for example, a CPU or a system on chip (SoC), and control functions of an application layer and additional layer of the smartphone **2500**. The memory **2502** includes an RAM and an ROM, and stores a program that is executed by the processor **2501** and data. The storage **2503** may include a storage medium such as a semiconductor memory and a hard disk. The external connection interface **2504** is an interface for connecting an external device (such as a memory card and a Universal Serial Bus (USB) device) to the smartphone **2500**.

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The image capturing device **2506** includes an image sensor (such as a charge coupled device (CCD) and a complementary metal oxide semiconductor (CMOS)), and generates a captured image. The sensor **2507** may include a group of sensors, such as a measurement sensor, a gyro sensor, a geomagnetic sensor and an acceleration sensor. The microphone **2508** converts sound inputted to the smartphone **2500** into an audio signal. The input device **2509** includes, for example, a touch sensor configured to detect touch onto a screen of the display device **2510**, a keypad, a keyboard, a button, or a switch, and receives an operation or information inputted by a user. The display device **2510** includes a screen such as a liquid crystal display (LCD) and an organic light-emitting diode (OLED) display, and displays an output image of the smartphone **2500**. The speaker **2511** is configured to convert an audio signal outputted from the smartphone **2500** into sound.

The wireless communication interface **2512** supports any cellular communication scheme such as LTE and LTE-Advanced, and performs wireless communication. The wireless communication interface **2512** may generally include, for example, a baseband (BB) processor **2513** and a radio frequency (RF) circuit **2514**. The BB processor **2513** may perform, for example, encoding/decoding, modulation/demodulation, and multiplexing/demultiplexing, and perform various types of signal processing for wireless communication. The RF circuit **2514** may include, for example, a mixer, a filter and an amplifier, and transmits and receives a wireless signal via the antenna **2516**. The wireless communication interface **2512** may be a chip module having a BB processor **2513** and an RF circuit **2514** integrated thereon. The wireless communication interface **2512** may include multiple BB processors **2513** and multiple RF circuits **2514**, as shown in FIG. 8. Although an example in which the wireless communication interface **2512** includes multiple BB processors **2513** and multiple RF circuits **2514** is shown in FIG. 8, the wireless communication interface **2512** may include a single BB processor **2513** and a single RF circuit **2514**.

Furthermore, in addition to a cellular communication scheme, the wireless communication interface **2512** may support another type of wireless communication scheme such as a short-range wireless communication scheme, a near field communication scheme, and a wireless local area network LAN scheme. In this case, the wireless communication interface **2512** may include a BB processor **2513** and an RF circuit **2514** for each wireless communication scheme.

Each of the antenna switches **2515** switches connection destinations of the antennas **2516** among multiple circuits (such as circuits for different wireless communication schemes) included in the wireless communication interface **2512**.

Each of the antennas **2516** includes a single or multiple antenna elements (such as multiple antenna elements included in an MIMO antenna), and is used for the wireless communication interface **2512** to transmit and receive a wireless signal. As shown in FIG. 8, the smartphone **2500** may include multiple antennas **2516**. Although FIG. 8 shows an example in which the smartphone **2500** includes multiple antennas **2516**, the smartphone **2500** may include a single antenna **2516**.

In addition, the smartphone **2500** may include an antenna **2516** for each wireless communication scheme. In this case, the antenna switch **2515** may be omitted from the configuration of the smartphone **2500**.

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The processor **2501**, the memory **2502**, the storage device **2503**, the external connection interface **2504**, the image capturing device **2506**, the sensor **2507**, the microphone **2508**, the input device **2509**, the display device **2510**, the speaker **2511**, the wireless communication interface **2512**, and the auxiliary controller **2519** are connected to each other via the bus **2517**. The battery **2518** supplies power to the respective blocks of the smartphone **2500** as shown in FIG. 8 via feeder lines which are partially shown as dashed lines in the figure. The auxiliary controller **2519** manipulates a minimum necessary function of the smartphone **2500** in a sleep mode, for example.

In the smartphone **2500** shown in FIG. 8, the transceiver device of the wireless communication device at the user equipment side according to the embodiment of the present disclosure may be implemented by the wireless communication interface **2512**. At least a part of the functions of the processing circuitry and/or respective units of the electronic device or the wireless communication device at the user equipment side according to the embodiment of the present disclosure may also be implemented by the processor **2501** or the auxiliary controller **2519**. For example, a part of the function of the processor **2501** may be implemented by the auxiliary controller **2519** so as to reduce power consumption of the battery **2518**. Further, the processor **2501** or the auxiliary controller **2519** may perform at least a part of the functions of the processing circuitry and/or respective units of the electronic device or wireless communication device at user equipment side according to the embodiment of the present disclosure by executing the program stored in the memory **2502** or the storage device **2503**.

[Exemplary Application Regarding a Base Station]

FIG. 9 is a block diagram showing an exemplary configuration of a gNB to which the technology according to the present disclosure may be applied. A gNB **2300** includes one or more antennas **2310** and a base station equipment **2320**. The base station equipment **2320** and each of the antenna **2310** may be connected with each other via a radio frequency (RF) cable.

Each of the antennas **2310** includes a single or multiple antenna elements (such as multiple antenna elements included in a Multi-Input Multi-Output (MIMO) antenna), and is configured for the base station equipment **2320** to transmit and receive a wireless signal. As shown in FIG. 9, the gNB **2300** may include multiple antennas **2310**. For example, the multiple antennas **2310** may be compatible with multiple frequency bands used by the gNB **2300**. Although FIG. 9 shows an example in which the gNB **2300** includes multiple antennas **2310**, the gNB **2300** may include a single antenna **2310**.

The base station equipment **2320** includes a controller **2321**, a memory **2322**, a network interface **2323**, and a wireless communication interface **2325**.

The controller **2321** may be, for example, a CPU or a DSP, and operate various functions of a higher layer of the base station equipment **2320**. For example, the controller **2321** generates a data packet based on data in a signal processed by the wireless communication interface **2325**, and transmits the generated packet via the network interface **2323**. The controller **2321** may bundle data from multiple baseband processors to generate a bundled packet and transmit the generated bundled packet. The controller **2321** may have a logical function of performing control such as radio resource control, radio bearer control, mobility management, admission control, and scheduling. The control may be performed in corporation with a nearby gNB or core network node. The memory **2322** includes an RAM and an ROM, and stores a

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program executed by the controller 2321 and various types of control data (such as a terminal list, transmission power data and scheduling data).

The network interface 2323 is a communication interface for connecting the base station equipment 2320 to a core network 2324. The controller 2321 may communicate with the core network node or another gNB via the network interface 2323. In this case, the gNB 2300 may be connected with the core network node or another gNB via a logic interface (such as an SI interface and an X2 interface). The network interface 2323 may also be a wired communication interface or a wireless communication interface for a wireless backhaul. In a case where the network interface 2323 is a wireless communication interface, the network interface 2323 may use a higher frequency band for wireless communication than a frequency band used by the wireless communication interface 2325.

The wireless communication interface 2325 supports any cellular communication scheme (such as Long Term Evolution (LTE) and LTE-Advanced), and provides wireless connection to a terminal located in a cell of the gNB 2300 via the antenna 2310. The wireless communication interface 2325 may generally include a BB processor 2326 and an RF circuit 2327. The BB processor 2326 may perform for example coding/decoding, modulation/demodulation and multiplexing/demultiplexing, and execute various types of signal processing of a layer (for example L1, Medium Access Control (MAC), Radio Link Control (RLC) and Packet Data Convergence Protocol (PDCP)). Instead of the controller 2321, the BB processor 2326 may have a part or all of the above-mentioned logic functions. The BB processor 2326 may be a memory storing communication control programs, or a module including a processor which is configured to execute the programs and a related circuit. Update of the programs may change the function of the BB processor 2326. The module may be a card or blade inserted into a slot of the base station equipment 2320. Alternatively, the module may be a chip mounted on a card or blade. The RF circuit 2327 may include, for example, a mixer, a filter and an amplifier, and transmits and receives a wireless signal via the antenna 2310.

As shown in FIG. 9, the wireless communication interface 2325 may include multiple BB processors 2326. For example, the multiple BB processors 2326 may be compatible with multiple frequency bands used by the gNB 2300. As shown in FIG. 9, the wireless communication interface 2325 may include multiple RF circuits 2327. For example, the multiple RF circuits 2327 may be compatible with multiple antenna elements. Although an example in which the wireless communication interface 2325 includes multiple BB processors 2326 and multiple RF circuits 2327 is shown in FIG. 9, the wireless communication interface 2325 may include a single BB processor 2326 and a single RF circuit 2327.

In the gNB 2300 shown in FIG. 9, the transceiver device of the wireless communication device at the base station side according to the embodiment of the present disclosure may be implemented by the wireless communication interface 2325. At least a part of the functions of the processing circuitry and/or respective units of the electronic device or the wireless communication device at the base station side according to the embodiment of the present disclosure may also be implemented by the controller 2321. For example, the controller 2321 may perform at least a part of the functions of the processing circuitry and/or respective units of the electronic device or the wireless communication

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device at base station side according to the embodiment of the present disclosure by executing the program stored in the memory 2322.

[Exemplary Application in a Vehicle Navigation Device]

FIG. 10 is a block diagram showing an exemplary configuration of a vehicle navigation equipment 2120 to which the technology according to the present disclosure may be applied. The vehicle navigation device 2120 includes a processor 2121, a memory 2122, a global positioning system (GPS) module 2124, a sensor 2125, a data interface 2126, a content player 2127, a storage medium interface 2128, an input device 2129, a display device 2130, a speaker 2131, a wireless communication interface 2133, one or more antenna switches 2136, one or more antennas 2137, and a battery 2138.

The processor 2121 may be, for example, a CPU or a SoC, and control a navigation function and another function of the vehicle navigation device 2120. The memory 2122 includes an RAM and an ROM, and stores a program that is executed by the processor 2121 and data.

The GPS module 2124 uses a GPS signal received from a GPS satellite to measure a position (such as a latitude, a longitude, and an altitude) of the vehicle navigation device 2120. The sensor 2125 may include a group of sensors such as a gyro sensor, a geomagnetic sensor, and an air pressure sensor. The data interface 2126 is connected to, for example, an in-vehicle network 2141 via a terminal that is not shown, and acquires data generated by the vehicle (such as vehicle speed data).

The content player 2127 reproduces content stored in a storage medium (such as a CD and a DVD) inserted into the storage medium interface 2128. The input device 2129 includes, for example, a touch sensor configured to detect touch on a screen of the display device 2130, a button, or a switch, and receives an operation or information inputted by a user. The display device 2130 includes a screen such as a LCD or an OLED display, and displays an image of the navigation function or content that is reproduced. The speaker 2131 outputs a sound of the navigation function or content that is reproduced.

The wireless communication interface 2133 supports any cellular communication scheme such as LTE and LTE-Advanced, and performs wireless communication. The wireless communication interface 2133 may generally include a BB processor 2134 and an RF circuit 2135. The BB processor 2134 may perform, for example, encoding/decoding, modulation/demodulation, and multiplexing/demultiplexing, and perform various types of signal processing for wireless communication. The RF circuit 2135 may include, for example, a mixer, a filter and an amplifier, and transmits and receives a wireless signal via the antenna 2137. The wireless communication interface 2133 may be a chip module having a BB processor 2134 and an RF circuit 2135 integrated thereon. The wireless communication interface 2133 may include multiple BB processors 2134 and multiple RF circuits 2135, as shown in FIG. 10. Although an example in which the wireless communication interface 2133 includes multiple BB processors 2134 and multiple RF circuits 2135 is shown in FIG. 10, the wireless communication interface 2133 may include a single BB processor 2134 and a single RF circuit 2135.

Furthermore, in addition to a cellular communication scheme, the wireless communication interface 2133 may support another type of wireless communication scheme such as a short-range wireless communication scheme, a near field communication scheme, and a wireless LAN scheme. In this case, the wireless communication interface

2133 may include a BB processor **2134** and an RF circuit **2135** for each wireless communication scheme.

Each of the antenna switches **2136** switches connection destinations of the antennas **2137** among multiple circuits (such as circuits for different wireless communication schemes) included in the wireless communication interface **2133**.

Each of the antennas **2137** includes a single or multiple antenna elements (such as multiple antenna elements included in an MIMO antenna), and is used for the wireless communication interface **2133** to transmit and receive a wireless signal. As shown in FIG. 10, the vehicle navigation device **2120** may include multiple antennas **2137**. Although FIG. 10 illustrates an example in which the vehicle navigation device **2120** includes multiple antennas **2137**, the vehicle navigation device **2120** may include a single antenna **2137**.

In addition, the vehicle navigation device **2120** may include an antenna **2137** for each wireless communication scheme. In this case, the antenna switches **2136** may be omitted from the configuration of the vehicle navigation device **2120**.

The battery **2138** supplies power to the respective blocks of the vehicle navigation device **2120** shown in FIG. 10 via feeder lines that are partially shown as dashed lines in the Figure. The battery **2138** accumulates power supplied from a vehicle.

In the vehicle navigation device **2120** shown in FIG. 10, the transceiver device or the transceiver unit of the wireless communication device according to the embodiment of the present disclosure may be implemented by the wireless communication interface **2133**. At least a part of the functions of the processing circuitry and/or respective units of the electronic device or the wireless communication device according to the embodiment of the present disclosure may also be implemented by the processor **2121**.

The technology of the present disclosure may also be implemented as an in-vehicle system (or a vehicle) **2140** including one or more of the vehicle navigation device **2120**, an in-vehicle network **2141** and a vehicle module **2142**. The vehicle module **2142** generates vehicle data (such as a vehicle speed, an engine speed, and fault information), and outputs the generated data to the in-vehicle network **2141**.

In the above description of specific embodiments of the present disclosure, features described and/or illustrated for one embodiment may be used in one or more other embodiments in the same or similar manner, or may be combined with features in other embodiments, or may replace features in other embodiments.

It is be noted that, terms “including/comprising” used herein refer to existing of features, elements, steps or components, but existing or adding of one or more other features, elements, steps or components is not excluded.

In the above embodiments and examples, reference numerals consist of numbers are used to represent steps and/or units. Those skilled in the art should understand that the reference numerals are used to facilitate describing and drawing, and are not intended to indicate an order or limitation in any way.

In addition, the method according to the present disclosure is not limited to be performed in the chronological order described herein, and may be performed in other chronological order, in parallel or independently. Therefore, the order in which the method is performed described herein does not limit the technical scope of the present disclosure.

Although the present disclosure is disclosed by the description of specific embodiments of the present disclosure

sure above, it should be understood that all the embodiments and examples described above are only exemplary and are not intended to limit. Various modifications, improvements and equivalents may be made to the present disclosure by those skilled in the art within the scope and spirit of the attached claims. These modifications, improvements or equivalents shall also fall within the protection scope of the present disclosure.

In addition, the following solutions are further provided according to embodiments of present disclosure.

(1) An electronic device for wireless communication, comprising processing circuitry configured to:

perform control to transmit data and information on retransmission of the data to target user equipment via a sidelink; and

control an operation for retransmission of the data.

(2) The electronic device according to (1), wherein the information comprises a time gap between initial transmission and retransmission of the data; and

the operation comprises performing retransmission of the data according to the time gap.

(3) The electronic device according to (1), wherein the information comprises time domain position information of a feedback to the data; and

the operation comprises receiving feedback information of the target user equipment to the data based on the time domain position information, and performing retransmission of the data based on the feedback information.

(4) The electronic device according to (3), wherein the information further comprises information indicating a retransmission unit; and

the operation comprises receiving a feedback to the data and performing retransmission of the data in units of transmission block or code block group.

(5) The electronic device according to (3), wherein the time domain position information comprises:

one or more time domain positions for receiving the feedback information; and/or

one or more time windows for receiving the feedback information.

(6) The electronic device according to (5), wherein the time domain position information comprises a plurality of time domain positions for receiving the feedback information, and the operation comprises:

receiving the feedback information successively at the plurality of time domain positions until the feedback information is received; and

performing retransmission if the feedback information is not received at any of the plurality of time domain positions.

(7) The electronic device according to (5), wherein the time domain position information comprises a plurality of time windows for receiving the feedback information, and the operation comprises:

receiving the feedback information successively in the plurality of time windows until the feedback information is received; and

performing retransmission if the feedback information is not received in any of the plurality of time windows.

(8) The electronic device according to (5), wherein the time domain position information comprises the time domain positions and the time windows for receiving the feedback information, and the operation comprises:

receiving the feedback information successively at the time domain positions and in the time windows until the feedback information is received; and

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performing retransmission if the feedback information is not received at any of the time domain positions or in any of the time windows.

(9) The electronic device according to (1), wherein the information comprises one or more of:

resource reservation, source user equipment identification, target user equipment identification, frequency domain position(s) for initial transmission and retransmission, frequency domain position(s) for initial transmission and feedback transmission, modulation and coding scheme, retransmission index, code block group transmission information, and service priority of the data.

(10) The electronic device according to (3), wherein the operation further comprises:

decoding the received feedback information; and determining whether the decoded feedback information is for the data.

(11) The electronic device according to (4), wherein the operation comprises performing feedback to the data and retransmission of the data in units of code block group, and the feedback information comprises codebook information.

(12) The electronic device according to (3), wherein the feedback information comprises information on retransmission resources for the data; and

the operation further comprises determining retransmission resources for the data according to the information on retransmission resources.

(13) A wireless communication method, comprising: transmitting data and information on retransmission of the data to target user equipment via a sidelink; and performing an operation for retransmission of the data.

(14) An electronic device for wireless communication, comprising processing circuitry configured to:

perform control to receive data and information on retransmission of the data from source user equipment via a sidelink; and

control an operation for retransmission of the data.

(15) The electronic device according to (14), wherein the information comprises a time gap between initial transmission and retransmission of the data; and

the operation comprises receiving retransmission of the data according to the time gap.

(16) The electronic device according to (14), wherein the information comprises time domain position information of a feedback to the data; and

the operation comprises transmitting the feedback information to the data to the source user equipment based on the time domain position information.

(17) The electronic device according to (16), wherein the information further comprises information indicating a retransmission unit; and

the operation comprises performing the feedback to the data in units of transmission block or code block group.

(18) The electronic device according to (16), wherein the time domain position information comprises one or more time domain positions for transmitting the feedback information and/or one or more time windows for transmitting the feedback information; and

the operation comprises transmitting the feedback information at one of the one or more time domain positions or in one of the one or more time windows.

(19) The electronic device according to (16), wherein the information comprises a service priority of the data; and the operation comprises:

transmitting the feedback information and receiving other data based on the service priority; and/or

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transmitting the feedback information for a plurality of links based on the service priority.

(20) The electronic device according to (19), wherein the operation further comprises:

selecting, according to the priority, a link for which the feedback information is to be transmitted, in a case where the number of the plurality of links exceeds a predetermined threshold.

(21) The electronic device according to (19), wherein the operation further comprises distinguishing the feedback information for the plurality of links from each other by:

comprising identification information of corresponding source user equipment in the feedback information; and/or

scrambling the feedback information with a codebook.

(22) The electronic device according to (16), wherein the operation further comprises:

comprising information on a retransmission resource for the data in the feedback information, and receiving retransmission of the data based on the retransmission resource.

(23) A wireless communication method, comprising: receiving data and information on retransmission of the data from source user equipment via a sidelink; and performing an operation for retransmission of the data.

(24) An electronic device for wireless communication, comprising processing circuitry configured to:

set sidelink control information for indicating information on data retransmission performed from source user equipment to target user equipment via a sidelink; and perform control to notify the set sidelink information to the source user equipment,

wherein the information on data retransmission comprises a time gap between initial transmission and retransmission of the data by the source user equipment, or time domain position information of a feedback to the data by the target user equipment.

(25) A wireless communication method, comprising: setting sidelink control information for indicating information on data retransmission performed from source user equipment to target user equipment via a sidelink; and

notifying the set sidelink control information to the source user equipment,

wherein the information on data retransmission comprises a time gap between initial transmission and retransmission of the data by the source user equipment, or time domain position information of a feedback to the data by the target user equipment.

(26) A computer readable medium comprising executable instructions that, when executed by an information processing apparatus, cause the information processing apparatus to implement the method according to any one of (13), (23) and (25).

The invention claimed is:

1. An electronic device in a wireless communication system, the electronic device comprising processing circuitry configured to:

transmit data to a target user equipment via a sidelink;

transmit information on a retransmission of the data to the target user equipment,

wherein the information on the retransmission of the data indicates time and frequency resources for receiving feedback information relating to feedback of the data,

wherein the time and frequency resources include a first time in a first time slot and a second time in a second time slot sequential to the first time slot, and

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wherein the first time is one of a time domain position or a time window within the first time slot, and the second time is one of a time domain position or a time window within the second time slot;
 receive the feedback information in the time and frequency resources of the first and second time slots; and perform a retransmission of the data based on the feedback information.

2. The electronic device according to claim 1, wherein the information on the retransmission of the data further comprises a time gap between an initial transmission of the data and the retransmission of the data; and the circuitry is further configured to perform the retransmission of the data according to the time gap.

3. The electronic device according to claim 1, wherein the information on the retransmission of the data includes information indicating a retransmission unit; and the circuitry is further configured to receive the feedback information and perform the retransmission of the data in units of a transmission block (TB) or a code block group (CBG) based on the information indicating the retransmission unit.

4. The electronic device according to claim 1, wherein the information on the retransmission of the data further comprises one or more of:
 a resource reservation, a source user equipment identification, a target user equipment identification, a frequency domain position for an initial transmission and the retransmission of the data, a frequency domain position for the initial transmission and a transmission of the feedback information, a modulation and coding scheme, a retransmission index, code block group (CBG) transmission information, or a service priority of the data.

5. The electronic device according to claim 1, wherein the circuitry is further configured to:
 perform the retransmission of the data only in a case that the feedback information comprises NACK information and not in a case that the feedback information comprises ACK information or no feedback information.

6. The electronic device according to claim 1, wherein the information on the retransmission of the data includes information indicating a retransmission unit, wherein the circuitry is further configured to receive the feedback information and determine whether the information indicating a retransmission unit indicates that the retransmission of the data is to be in units of a transmission block (TB) or in units of a code block group (CBG),
 upon determining that the retransmission of the data is to be in units of the TB, perform the retransmission of the data of a specific TB corresponding to a HARQ NACK, and
 upon determining that the retransmission of the data is to be in units of the CBG, receive information about a feedback codebook and perform the retransmission of the data of a CBG upon determining, via the feedback codebook, that feedback for the CBG is NACK.

7. The electronic device according to claim 1, wherein the feedback information indicates information on retransmission resources for data; and the circuitry is further configured to:
 perform the retransmission of the data by utilizing the retransmission resources.

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8. The electronic device according to claim 1, wherein the first time is the time domain position within the first time slot, and the second time is the time window within the second time slot, or
 wherein the first time is the time window within the first time slot, and the second time is the time domain position within the second time slot.

9. An electronic device in a wireless communication system, the electronic device comprising processing circuitry configured to:
 receive data from a source user equipment via a sidelink;
 receive information on a retransmission of the data from the source user equipment;
 wherein the information on the retransmission of the data indicates time and frequency resources for transmitting feedback information relating to feedback of the data, wherein the time and frequency resources include a first time in a first time slot and a second time in a second time slot sequential to the first time slot, and
 wherein the first time is one of a time domain position or a time window within the first time slot, and the second time is one of a time domain position or a time window within the second time slot;
 transmit the feedback information multiple times in the time and frequency resources of the first and second time slots; and
 receive a retransmission of the data based on the feedback information.

10. The electronic device according to claim 9, wherein the information on the retransmission of the data further comprises a time gap between an initial transmission of the data and the retransmission of the data; and the circuitry is further configured to receive the retransmission of the data according to the time gap.

11. The electronic device according to claim 9, wherein the information on the retransmission of the data includes information indicating a retransmission unit; and the circuitry is further configured to transmit the feedback information and receive the retransmission of the data in units of a transmission block (TB) or a code block group (CBG) based on the information indicating the retransmission unit.

12. The electronic device according to claim 9, wherein the information on the retransmission of the data further comprises a service priority of the data; and the circuitry is further configured to:
 transmit the feedback information and receive other data based on the service priority; or
 transmit the feedback information for a plurality of links based on the service priority.

13. The electronic device according to claim 12, wherein the circuitry is further configured to:
 select according to the service priority, a link for which the feedback information is to be transmitted, in a case where a number of the plurality of links exceeds a predetermined threshold.

14. The electronic device according to claim 9, wherein the circuitry is further configured to:
 distinguish the feedback information for the plurality of links from each other by at least one of:
 including identification information of a corresponding source user equipment in the feedback information; or
 scrambling the feedback information with a scrambling codebook.

15. The electronic device according to claim 9, wherein the feedback information indicates information on retransmission resources for the data; and

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the circuitry is further configured to:
receive the retransmission of the data by utilizing the
retransmission resources.

16. The electronic device according to claim 9,
wherein the first time is the time domain position within
the first time slot, and the second time is the time
window within the second time slot, or
wherein the first time is the time window within the first
time slot, and the second time is the time domain
position within the second time slot.

17. A method performed by a wireless communication
device, the method comprising:

transmitting data to a target user equipment via a sidelink;
transmitting information on a retransmission of the data to
the target user equipment,

wherein the information on the retransmission of the data
indicates time and frequency resources for receiving
feedback information relating to feedback of the data,
wherein the time and frequency resources include a first
time in a first time slot and a second time in a second
time slot sequential to the first time slot, and

wherein the first time is one of a time domain position or
a time window within the first time slot, and the second
time is one of a time domain position or a time window
within the second time slot;

receiving the feedback information multiple times in the
time and frequency resources of the first and second
time slots; and

performing a retransmission of the data based on the
feedback information.

18. The method according to claim 17, wherein
the information on the retransmission of the data further
comprises a time gap between an initial transmission of
the data and the retransmission of the data; and
the method further comprising:
performing the retransmission of the data according to the
time gap.

19. The method according to claim 17, wherein
the information on the retransmission of the data includes
information indicating a retransmission unit; and
the method further comprising:
receiving the feedback information and performing the
retransmission of the data in units of a transmission

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block (TB) or a code block group (CBG) based on the
information indicating the retransmission unit.

20. The method according to claim 17,
wherein the information on the retransmission of the data
includes information indicating a retransmission unit,
wherein the method further comprises:

receiving the feedback information and determining
whether the information indicating a retransmission
unit indicates that the retransmission of the data is to be
in units of a transmission block (TB) or in units of a
code block group (CBG),

upon determining that the retransmission of the data is to
be in units of the TB, performing the retransmission of
the data of a specific TB corresponding to a HARQ
NACK, and

upon determining that the retransmission of the data is to
be in units of the CBG, receiving information about a
feedback codebook and performing the retransmission
of the data of a CBG upon determining, via the feed-
back codebook, that feedback for the CBG is NACK.

21. The method according to claim 17, further compris-
ing:

performing the retransmission of the data only in a case
that the feedback information comprises NACK infor-
mation and not in a case that the feedback information
comprises ACK information or no feedback informa-
tion is received.

22. The method according to claim 17, wherein
the feedback information indicates information on
retransmission resources for data; and
the method further comprises:
performing the retransmission of the data by utilizing the
retransmission resources.

23. The method according to claim 17,
wherein the first time is the time domain position within
the first time slot, and the second time is the time
window within the second time slot, or
wherein the first time is the time window within the first
time slot, and the second time is the time domain
position within the second time slot.

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